

THE FIRST MISSING COTRANSVERSE MAP OCCURS IN DIMENSION FOUR

PHILIPPE GAUCHER

ABSTRACT. We correct the proof of Theorem 3.1.16 of *Combinatorics of labelling in higher-dimensional automata*. In the notation of *Towards a theory of natural directed paths*, let $\overline{\square}$ be the category of cubes generated by the coface maps δ_i^α , the symmetry maps σ_i , and the cotransverse degeneracy maps γ_i . The map proposed in the 2010 paper is a composite of two comparators and therefore belongs to $\overline{\square}$. We prove

$$\overline{\square}([3], [3]) = \widehat{\square}_S([3], [3]) \quad \text{and} \quad \overline{\square}([4], [4]) \subsetneq \widehat{\square}_S([4], [4]).$$

The equality in dimension three is obtained from the incidence cycle C_6 between the vertices of ranks one and two of the 3-cube. The strict inclusion in dimension four is witnessed by an explicit map constructed from the path graph on four vertices. The Lean formalization is available here: <https://github.com/pgs0042/Cotransverse-Maps>.

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1. CONVENTIONS AND RECALLED DEFINITIONS

We use only the terminology and notation of [3]. Thus $[n] = \{0 < 1\}^n$, the box category $\square$ is generated by the coface maps δ_i^α , and a map $f: [m] \rightarrow [n]$ of $\mathbf{PoSet}^+$ is *cotransverse* when it takes every directed edge to a directed edge. Equivalently,

$$\overrightarrow{d}_1(x, y) = 1 \implies \overrightarrow{d}_1(f(x), f(y)) = 1.$$

The category of all cotransverse maps is denoted by $\widehat{\square}_S$. For $1 \leq i < n$, the symmetry and cotransverse degeneracy maps are

$$\begin{aligned} \sigma_i(x_1, \dots, x_n) &= (x_1, \dots, x_{i-1}, x_{i+1}, x_i, x_{i+2}, \dots, x_n), \\ \gamma_i(x_1, \dots, x_n) &= (x_1, \dots, x_{i-1}, \max(x_i, x_{i+1}), \min(x_i, x_{i+1}), x_{i+2}, \dots, x_n). \end{aligned}$$

The category generated by the maps δ_i^α , σ_i , and γ_i is denoted by $\overline{\square}$; see [3, Notation 2.15]. In particular, $\overline{\square} \subset \widehat{\square}_S$. Composition is written in the usual functional order: $uv = u \circ v$.

We identify a vertex of $[n]$ with a subset of $\{1, \dots, n\}$. For $A \subset \{1, \dots, n\}$, its characteristic vertex is denoted by χ_A . The rank of χ_A is $|A|$.

This work is AI-assisted.

1.1. Lemma (Rank preservation). *Every cotransverse map $f: [n] \rightarrow [n]$ preserves rank. In particular, it fixes 0_n and 1_n .*

Proof. Choose a maximal chain

$$0_n = x_0 < x_1 < \cdots < x_n = 1_n.$$

Every consecutive pair is a directed edge, so the vertices $f(x_0), \dots, f(x_n)$ form a directed path of length n in $[n]$. Consequently,

$$\text{rk}f(x_n) = \text{rk}f(x_0) + n.$$

Both ranks belong to $\{0, \dots, n\}$, hence $\text{rk}f(x_0) = 0$ and $\text{rk}f(x_n) = n$. Every vertex of rank k lies on a maximal chain and is the k -th vertex of that chain. The same argument therefore gives $\text{rk}f(x) = k$. $\square$

1.2. Proposition. *For $0 \leq n \leq 2$, one has*

$$\overline{\square}([n], [n]) = \widehat{\square}_S([n], [n]).$$

Proof. The cases $n = 0, 1$ are immediate from Lemma 1.1. For $n = 2$, a cotransverse endomorphism fixes 0_2 and 1_2 and is determined by an arbitrary map from the two rank-one vertices to themselves. The four possibilities are realized respectively by

$$\text{id}_{[2]}, \quad \sigma_1, \quad \gamma_1, \quad \sigma_1\gamma_1.$$

$\square$

2. THE PROPOSED THREE-DIMENSIONAL COUNTEREXAMPLE

Let $f: [3] \rightarrow [3]$ be the map used in the proof of [2, Theorem 3.1.16]. Its values, as specified there, are

x	$f(x)$
$(0, 0, 0)$	$(0, 0, 0)$
$(1, 0, 0), (0, 1, 0), (0, 0, 1)$	$(0, 0, 1)$
$(1, 1, 0), (0, 1, 1)$	$(0, 1, 1)$
$(1, 0, 1)$	$(1, 0, 1)$
$(1, 1, 1)$	$(1, 1, 1)$

Equivalently,

$$(1) \quad f(x_1, x_2, x_3) = (\min(x_1, x_3), \min(x_2, \max(x_1, x_3)), \max(x_1, x_2, x_3)).$$

For $1 \leq i < j \leq n$, let $c_{ij}: [n] \rightarrow [n]$ be the comparator which replaces (x_i, x_j) by $(\min(x_i, x_j), \max(x_i, x_j))$ and leaves all other coordinates unchanged. The adjacent comparator is

$$c_{i,i+1} = \sigma_i\gamma_i.$$

Every c_{ij} is therefore in $\overline{\square}$: it is a conjugate of an adjacent comparator by symmetry maps.

2.1. Proposition. *The map f of (1) belongs to $\overline{\square}([3], [3])$. More precisely,*

$$f = c_{23}c_{13} = \sigma_2\gamma_2\sigma_2\sigma_1\gamma_1\sigma_2.$$

Proof. The first comparator gives

$$c_{13}(x_1, x_2, x_3) = (\min(x_1, x_3), x_2, \max(x_1, x_3)).$$

Applying c_{23} yields exactly the right-hand side of (1). Moreover,

$$c_{13} = \sigma_2(\sigma_1\gamma_1)\sigma_2 \quad \text{and} \quad c_{23} = \sigma_2\gamma_2,$$

which proves the displayed word in the generators of $\overline{\square}$. $\square$

The erroneous step in the old proof can now be pinpointed. It obtains the unique factorizations

$$f\delta_1^0 = \delta_1^0(\sigma_1\gamma_1), \quad f\delta_1^1 = \delta_3^1\sigma_1,$$

and treats the inequality $\sigma_1\gamma_1 \neq \sigma_1$ between the two residual maps $[2] \rightarrow [2]$ as an obstruction to expressing f as a composite of the σ_i and γ_i . This implication is false: both displayed identities hold for the generated map of Proposition 2.1. More generally, when a coface map is moved through a word of symmetry and cotransverse degeneracy maps, both its position and the resulting lower-dimensional map may depend on $\alpha \in \{0, 1\}$. Thus the map in the 2010 paper is cotransverse, but it is not a counterexample to equality with $\overline{\square}$.

3. ALL COTRANSVERSE ENDOMORPHISMS OF THE THREE-DIMENSIONAL CUBE

We recall the small amount of graph terminology used below. A graph homomorphism is a map on vertices which takes adjacent vertices to adjacent vertices; an endomorphism is a graph homomorphism from a graph to itself. The cycle graph with six vertices is denoted by C_6 . These are standard definitions; see, for example, [4, Chapter 1]. Describing an endomorphism of a cycle by the closed walk which is the image of its cyclically ordered vertex set is standard in the study of cycle endomorphism monoids; compare [1, Proposition 1.1 and Proposition 1.9].

Let Γ_3 be the incidence graph whose vertices are the three rank-one vertices and the three rank-two vertices of [3], with an edge for each containment. It is the cycle C_6 , and its two parts are the two ranks. By Lemma 1.1, restriction induces a bijection

$$(2) \quad \widehat{\square}_S([3], [3]) \xrightarrow{\cong} \text{End}_{\text{bip}}(C_6),$$

where the right-hand side consists of the endomorphisms preserving the two parts of the bipartition. Indeed, restriction gives such an endomorphism. Conversely, a bipartition-preserving endomorphism of the incidence graph, together with the forced assignments $0_3 \mapsto 0_3$ and $1_3 \mapsto 1_3$, preserves every cover relation and hence defines a cotransverse map.

We now give a structural enumeration and generation of the right-hand side of (2). Number the vertices of C_6 cyclically by $v_0, \dots, v_5$, with the even vertices in one part. If h preserves the bipartition, then $h(v_0)$ has three possible values. There are signs $\varepsilon_0, \dots, \varepsilon_5 \in \{-1, 1\}$ such that

$$h(v_{i+1}) = v_{j_i + \varepsilon_i} \quad \text{whenever} \quad h(v_i) = v_{j_i},$$

with indices taken modulo 6. Closure of the walk says

$$(3) \quad \sum_{i=0}^5 \varepsilon_i \equiv 0 \pmod{6}.$$

Thus either all signs are equal, or three signs are 1 and three are -1 . It follows at once that

$$(4) \quad |\widehat{\square}_S([3], [3])| = 3 \left(2 + \binom{6}{3} \right) = 66.$$

For completeness, we identify the generators geometrically. The coordinate permutations induce all six automorphisms of C_6 which preserve its bipartition: $\text{Aut}_{\text{bip}}(C_6)$ has order six, and its faithful action on the three rank-one vertices identifies it with the symmetric group $\mathfrak{S}_3$. To check the relevant folding explicitly, take

$$\begin{aligned} v_0 &= \chi_{\{1\}}, & v_1 &= \chi_{\{1,2\}}, & v_2 &= \chi_{\{2\}}, \\ v_3 &= \chi_{\{2,3\}}, & v_4 &= \chi_{\{3\}}, & v_5 &= \chi_{\{1,3\}}. \end{aligned}$$

In this cyclic numbering, the restriction of γ_1 is $(0, 1, 0, 5, 4, 5)$. If $r(i) = i + 2 \pmod{6}$, then this tuple is $r^{-1}ar$, where

$$(5) \quad a = (0, 1, 2, 3, 2, 1),$$

where the tuple records the successive images of $v_0, \dots, v_5$. The rotation r is induced by the cyclic permutation of the three coordinates. Thus a belongs to the monoid induced by $\widehat{\square}([3], [3])$. Three further foldings in the monoid generated by a and the bipartition-preserving automorphisms are

$$(6) \quad b = ara = (2, 3, 2, 1, 2, 3),$$

$$(7) \quad d = ar^2a = (2, 1, 0, 1, 0, 1),$$

$$(8) \quad c = (ar^2)b = (0, 1, 0, 1, 0, 1).$$

3.1. Lemma. *Every bipartition-preserving endomorphism of C_6 is generated by $\text{Aut}_{\text{bip}}(C_6)$ and a .*

Proof. The two constant-sign words in (3) give precisely the bipartition-preserving automorphisms. Consider a word with three plus signs and three minus signs, and let $S = \{i : \varepsilon_i = 1\} \subset \mathbb{Z}/6$. Precomposition by a rotation of the domain gives $S \mapsto S + 2$. A reflection of the target gives $S \mapsto S^c$, while a reflection of the domain gives $S \mapsto (-S - 1)^c$; hence the operation $S \mapsto -S - 1$ is available as well. These formulae follow directly by following the oriented edge from v_i to v_{i+1} .

The resulting action has four orbits, represented by

$$\underbrace{+++---}_a, \quad \underbrace{+--++-}_b, \quad \underbrace{--+-++}_d, \quad \underbrace{+-+-+-}_c.$$

Here is a complete check. Abbreviate $\{i, j, k\}$ to ijk . The plus-index sets in the four orbits are respectively

$$\begin{array}{c|l} a & 012, 234, 045, 345, 015, 123, \\ b & 034, 025, 124, 125, 134, 035, \\ d & 245, 014, 023, 013, 235, 145, \\ c & 024, 135. \end{array}$$

Indeed, each row is obtained from its first set by adding 2 modulo 6 and taking complements; the operation $S \mapsto -S - 1$ preserves the same row. The rows are disjoint and contain $6 + 6 + 6 + 2 = 20 = \binom{6}{3}$ sets, so no balanced word is missing. The four representatives are realized by (5)–(8). Finally, an even rotation of the target realizes each of the three possible values of $h(v_0)$. $\square$

3.2. Theorem. *There is no counterexample in dimension three:*

$$\overline{\square}([3], [3]) = \widehat{\square}_S([3], [3]).$$

Both sets have 66 elements.

Proof. The inclusion from left to right is part of the definition of $\overline{\square}$. Under (2), the symmetry maps generate $\text{Aut}_{\text{bip}}(C_6)$, and a conjugate of γ_1 restricts to a . Lemma 3.1 therefore shows that every cotransverse endomorphism of $[3]$ belongs to $\overline{\square}$. The cardinality statement is (4). $\square$

3.3. Corollary. *For every $0 \leq m \leq 3$ and every $n \geq m$, one has*

$$\overline{\square}([m], [n]) = \widehat{\square}_S([m], [n]).$$

Proof. By [3, Proposition 2.5], every cotransverse map $u: [m] \rightarrow [n]$ factors uniquely as

$$[m] \xrightarrow{v} [m] \xrightarrow{\delta} [n], \quad \delta \in \square, \quad v \in \widehat{\square}_S([m], [m]).$$

Proposition 1.2 and Theorem 3.2 imply that v belongs to $\overline{\square}$. The coface composite δ belongs to $\square \subset \overline{\square}$, and therefore so does $u = \delta v$. $\square$

4. A COTRANSVERSE ENDOMORPHISM IN DIMENSION FOUR

We first record an obstruction satisfied by every non-bijective map in $\overline{\square}([n], [n])$.

4.1. Definition. A map $u: [n] \rightarrow [n]$ *globally folds the pair* $\{p, q\} \subset \{1, \dots, n\}$ *if*

$$(9) \quad u(\chi_{A \cup \{p\}}) = u(\chi_{A \cup \{q\}}) \quad \text{for every } A \subset \{1, \dots, n\} \setminus \{p, q\}.$$

4.2. Lemma (First-fold obstruction). *Every non-bijective map of $\overline{\square}([n], [n])$ globally folds at least one pair.*

Proof. A composite from $[n]$ to $[n]$ cannot contain a coface map, since coface maps strictly raise dimension and all maps in a category of cubes have source dimension at most their target dimension. It is therefore a word in symmetry maps and cotransverse degeneracy maps. If it is non-bijective, the word contains a γ_i . Read the word in the order in which its factors are applied, and consider the first γ_i . All earlier factors form a coordinate permutation s . The map γ_i identifies

$$\chi_{B \cup \{i\}} \quad \text{and} \quad \chi_{B \cup \{i+1\}}$$

for every B disjoint from $\{i, i+1\}$. Put $p = s^{-1}(i)$ and $q = s^{-1}(i+1)$. The first part of the word thus identifies the two vertices in (9), and all later factors preserve this equality. Hence the full composite globally folds $\{p, q\}$. $\square$

Let P_4 denote the path graph with vertex set $\{1, 2, 3, 4\}$ and edge set

$$E(P_4) = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\}.$$

Define $F: [4] \rightarrow [4]$, in the subset notation, by

$$(10) \quad F(\chi_A) = \begin{cases} \chi_\emptyset, & |A| = 0, \\ \chi_{\{1\}}, & |A| = 1, \\ \chi_{\{1,2\}}, & |A| = 2 \text{ and } A \in E(P_4), \\ \chi_{\{1,3\}}, & |A| = 2 \text{ and } A \notin E(P_4), \\ \chi_{\{1,2,3\}}, & |A| = 3, \\ \chi_{\{1,2,3,4\}}, & |A| = 4. \end{cases}$$

4.3. Proposition. *The map F belongs to $\widehat{\square}_S([4], [4])$.*

Proof. For every cover $A \subset A \cup \{i\}$, the corresponding values in (10) form one of the strict inclusions

$$\emptyset \subset \{1\} \subset \{1, 2\} \text{ or } \{1, 3\} \subset \{1, 2, 3\} \subset \{1, 2, 3, 4\}.$$

Thus F is strictly increasing and takes every directed edge to a directed edge. It is cotransverse by definition. $\square$

4.4. Proposition. *The map F does not belong to $\overline{\square}([4], [4])$.*

Proof. The map is non-bijective because all four rank-one vertices have the same image. We show that it globally folds no pair, contradicting Lemma 4.2 if it belonged to $\overline{\square}$.

The path P_4 has no pair of twins in the following precise sense: for every $p \neq q$, there is $k \notin \{p, q\}$ which is adjacent to exactly one of p, q . This can be read directly from

$$N(1) = \{2\}, \quad N(2) = \{1, 3\}, \quad N(3) = \{2, 4\}, \quad N(4) = \{3\}$$

where $N(i)$ is for the open neighbourhood of i , that is, the set of vertices adjacent to i . Choose such a k . Exactly one of $\{p, k\}$ and $\{q, k\}$ is an edge of P_4 . Formula (10) then gives

$$F(\chi_{\{p,k\}}) \neq F(\chi_{\{q,k\}}).$$

Taking $A = \{k\}$ in (9) proves that F does not globally fold $\{p, q\}$. Since this holds for every pair, the proof is complete. $\square$

4.5. Theorem. *The strict inclusion asserted in [2, Theorem 3.1.16] is true, but its first possible dimension is four:*

$$\overline{\square} \subsetneq \widehat{\square}_S.$$

More precisely,

$$\overline{\square}([3], [3]) = \widehat{\square}_S([3], [3]), \quad \overline{\square}([4], [4]) \subsetneq \widehat{\square}_S([4], [4]),$$

and the map F of (10) witnesses the second strict inclusion.

Proof. Corollary 3.3 proves that the two categories have the same maps in every hom-set whose source has dimension at most three. Propositions 4.3 and 4.4 give the strict inclusion in dimension four, which implies the strict inclusion of the small categories. $\square$

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UNIVERSITÉ PARIS CITÉ, CNRS, IRIF, F-75013, PARIS, FRANCE
URL: <https://www.irif.fr/~gaucher>